

Natural Language Processing (2-0-2-2-3)

Course Objectives

- Offers an in-depth overview of Text analytics topics including working with real world data
- Expose students to the Information Extraction problems and end to end Natural Language Generation problems as applications.
- Introduce students to the various Neural Network methods for Natural Language Processing.

Course Outcome

- Perform text analysis involving classification and clustering.
- Get technical insights to the state of the art framework in Natural language processing
- Understand business applications for Natural language processing
- Implement meaningful courses or research projects using current Natural Language Processing technology.

Prerequisites:

- Python for data science, supervised Machine learning, Deep Learning and Reinforcement Learning, Unsupervised Machine Learning & Recommendation Systems

Course Content

1. **Introduction to Text Analytics and Text Classification** : Text analytics framework, lexical and syntactic processing, generating word clouds, applying stemming techniques, removing stop words, and creating N-gram models. Bag-of-Words representation, Part-of-Speech (POS) tagging, Named Entity Recognition (NER), TF-IDF vectorization, parsing, and foundational word embedding and vectorization techniques like Word2Vec.

(Hands-on: performing text classification on a news dataset using TF-IDF and Logistic Regression.)

2. **Advanced Text Feature Representations and Word Embeddings**: understanding of word embeddings and feature representations for text. Learners will explore Word2Vec architectures like Skip-gram and CBOW, understand word embedding spaces, calculate vector similarity measures, and visualize word embeddings. Limitations of traditional feature-based models in NLP.

(Hands-on : train a Word2Vec model on a small text corpus and visualize similar words using cosine similarity.)

3. **Sequence-to-Sequence (Seq2Seq) Modeling**: sequence learning in NLP, fundamentals of Seq2Seq learning, understanding the encoder-decoder framework, and the practical applications of Seq2Seq models in text generation and language translation. Building Seq2Seq models using Bi-LSTM networks and highlights challenges in handling sequential data. As a

(Hands-on: implement a Bi-LSTM-based Seq2Seq model for a simple English-to-English sentence paraphrasing or translation task.

4. **Transformer-based Models and BERT** :NLP advancements, pre-transformer architectures and their limitations, a detailed explanation of the attention mechanism. Transformer architecture, understanding both encoder and decoder modules along with concepts like self-attention and multi-head attention. Introduction to BERT and its applications in NLP tasks.

(Hands-on: Fine-tune a pre-trained BERT model (such as `bert-base-uncased`) for sentiment classification on a movie reviews dataset.

Reference books

1. Mugan, J. (2017). Natural language text processing with Python: Hands-on NLP in Python using NLTK, spaCy, gensim, and scikit-learn.
2. In Loonycorn (Firm), & Packt Publishing,. (2017). From 0 to 1: Machine learning, NLP & Python : cut to the chase.